

3. (a) The first law of thermodynamics is given by:

$$\Delta U = Q - W$$

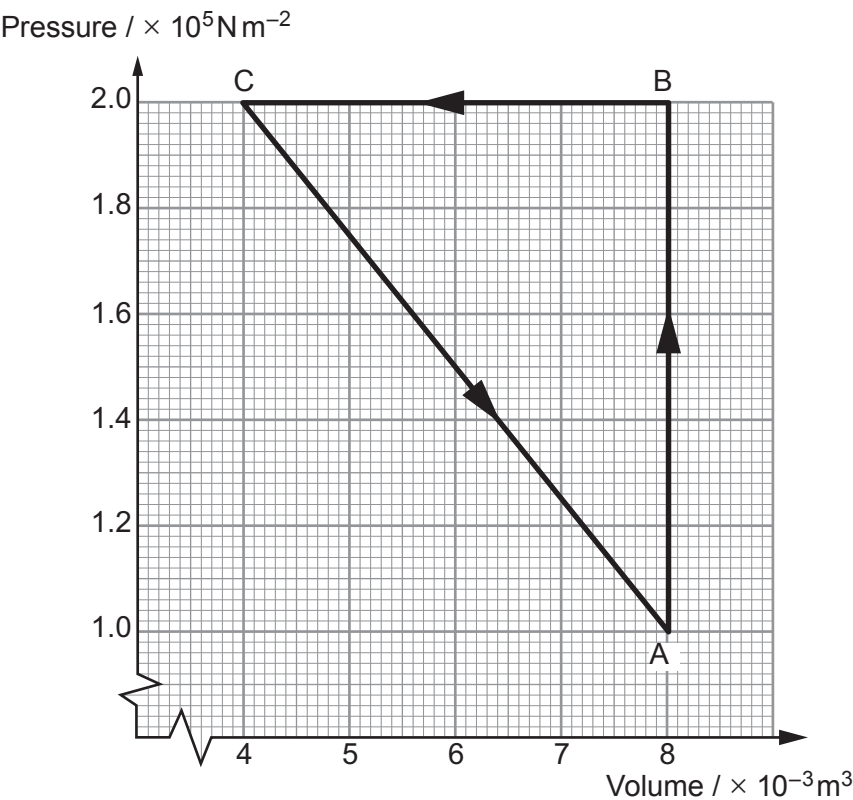
State what is represented by: [3]

ΔU

Q

W

(b) A fixed mass of gas is taken around the closed cycle $A \rightarrow B \rightarrow C \rightarrow A$.



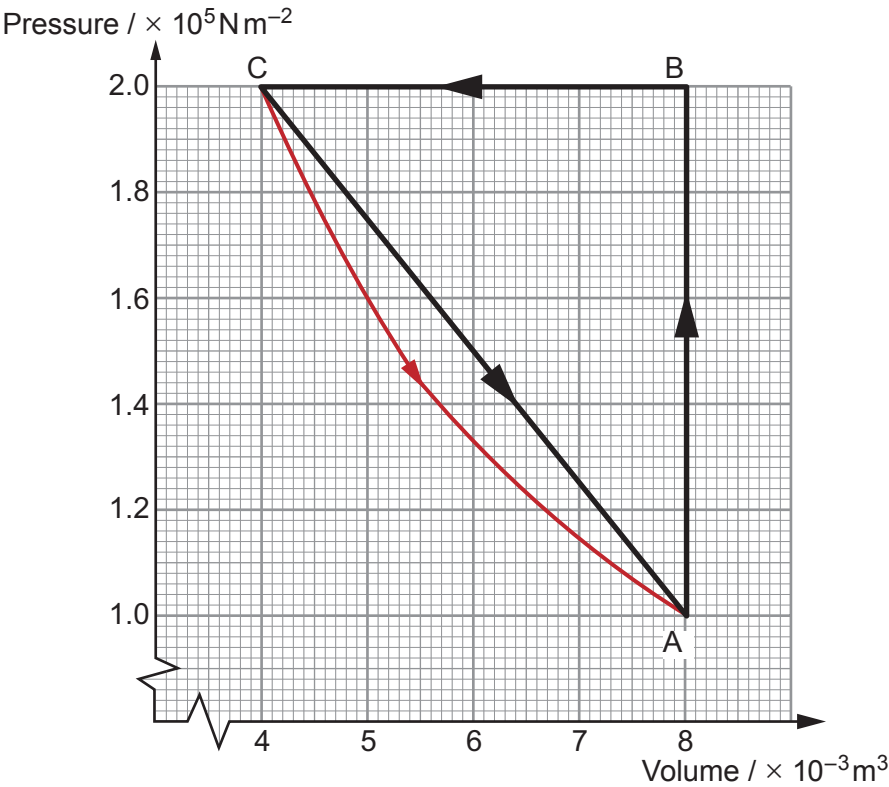
(i) Complete the table by describing each process in terms of pressure and volume. The description for process $A \rightarrow B$ is already inserted. For each process state if any work is done, and if so indicate if it is done *on* or *by* the gas. *No calculations are required.* [5]

Process	Description of process	Work done on / by gas (if any)
$A \rightarrow B$	Increase in pressure at constant volume	
$B \rightarrow C$		
$C \rightarrow A$		

(ii) Determine the net work done on the gas during the entire cycle.

[2] Examiner only

(c) The same mass of gas is taken around the cycle $A \rightarrow B \rightarrow C \rightarrow A$ for a second time. For this cycle the temperature is kept constant between C and A. The new path from C to A is shown in red on the graph below, together with the original path.



(i) Show that the temperature is constant along the new path from C to A.

[3]

(ii) Estimate the difference in the net work done in the two cycles.

[2]